

MoleCheck

A Phone App That Checks Moles With AI

Capstone Project Report

It looks at a photo of a mole and guesses whether it's probably harmless or worth getting checked — and the AI runs right on your phone.

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1. Summary

MoleCheck is a phone app that looks at a photo of one mole and gives its best guess: does it look harmless, or like something a doctor should check? An AI model does the guessing, and the whole thing runs **on the phone itself** — the photo never gets uploaded anywhere, so it's private and works without internet.

I tested it on 1,722 mole photos it had never seen before. It catches about **90% of the dangerous ones**, while correctly clearing about **75% of the harmless ones**. Its overall score for telling the two apart (called ROC-AUC, where 1.0 would be perfect) is **0.914**, which is pretty good.

One thing I want to be clear about: this is a school project, not a real medical tool. The whole app is built to tell people to go see a skin doctor, not to replace one.

2. Why I built this

Skin cancer is one of the most curable cancers if you catch it early, and one of the scariest if you don't. But not everyone can easily see a skin doctor, and most people have no idea which of their moles (if any) are worth worrying about. I wanted to build something that could give people a nudge to get a mole checked.

My goal was to do the whole thing myself, start to finish: train an AI on real mole photos, shrink it down so it runs on a phone with no server, and put it in an actual app you can use. So the project mixes AI, some model 'plumbing', and building a real mobile app.

3. The data

I trained the model on a public collection of mole photos called HAM10000 (from a big skin-image project called ISIC) — 11,720 photos, each labeled harmless or dangerous. Most of them are harmless, which matters later.

A sneaky mistake I avoided

The same mole is sometimes photographed a few times. If I let copies of the same mole end up in both my 'practice' pile and my 'test' pile, the model could basically memorize that exact mole and look way better than it really is. To stop that, I split the photos so **every picture of a given mole stays on one side only**.

Group	Moles	Photos	Dangerous %
Training	6,121	8,123	18.8%
Checking	1,309	1,726	18.1%
Testing	1,309	1,722	18.2%

4. How it works

The model is a kind of image AI called YOLO11 (5.4 million little numbers it tunes as it learns). It already knew a lot about images before I started, so I just fine-tuned it on moles — that's called

transfer learning, and it's why training only took about six minutes on my graphics card. I also flipped, rotated, and recolored the photos while training so the model doesn't get thrown off by different lighting or angles.

5. Results

Accuracy sounds like the obvious thing to measure, but it's sneaky here: since most moles are harmless, a lazy model that just says 'harmless' every single time already scores about **82%** — while catching zero cancers. So instead I care about two things: how many dangerous moles it catches, and how many harmless ones it correctly leaves alone.

What I measured	Score	What it means
ROC-AUC	0.914	How well it separates the two
Catch rate	90.1%	Dangerous moles it flagged
Clear rate	74.7%	Harmless moles it left alone

How careful to make it

Missing a cancer is way worse than a false alarm — a false alarm just means someone gets a mole checked that turns out fine. So I set the model to be **extra cautious**: it would rather flag too much than miss something.

6. Getting it running

To make the model run on a phone I had to convert it into a phone-friendly format. The normal way to do that didn't work on Windows, so I had to take a longer route through a couple of other formats. I didn't just trust that it still worked — I tested the converted model against the original on 32 photos, and they gave the **exact same answers every time**. Then I put it in a Flutter app (one set of code that works on both iPhone and Android), added a disclaimer you have to tap through first, and tested the whole thing on an Android phone.

7. Honest limitations

- It's not a real medical device — it's a school project and hasn't been tested for actual doctor use.
- It misses some cancers. Catching 90% means about 1 in 10 slips through, so a 'looks fine' result is NOT a green light.
- It learned on special close-up medical photos, so normal phone pics are harder for it and it'll probably do worse in real life.
- The photos it learned from don't include enough darker skin tones, which is a real fairness problem in skin AI.

That's exactly why the app always tells you to see a real doctor instead of trusting it.

8. What's next

- Train it on normal phone photos, not just medical close-ups.
- Test how well it works across different skin tones and fix the gaps.
- Let it say 'this photo is too blurry to judge' instead of guessing anyway.
- Finish the iPhone version and have some people actually try it.